

Deliverable: Change Report

Cohort 1, Team 8

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Introduction

Systematically, we interrogated each document comparing it to our original and the feedback given within the previous assessment, this enabled us to prioritise specific negative attributes to build upon. Firstly, we listed all improvement points by compiling them into a coherent and comprehensive list, assigning each task to a specific group member who had the most relevant experience alongside confidence. These changes were monitored by regular group discussion and the use of a Gantt Chart, this allowed us to visually display when tasks were scheduled to be complete. Throughout the development of our project we would regularly reflect with a peer group member to ensure we were meeting the objectives and improvements we initially defined.

Consistency was important throughout the project, we felt this was paramount due to its underlying nature to provide our group with a thorough and detailed culture. We ensured this was achieved by producing a routine, which consists of regular group meetings to evaluate progress. Within our earliest group meetings we looked at the current code repository, we felt it was extremely important we had an underpinning foundation of the code to ensure we could follow a similar design process ensuring the initial vision of the project hadn't been lost instead built upon. This goal was achieved by reading the pre-existing documentation associated with the code and mimicking the style and format.

Throughout the programming aspect of the project we were massively aided by Git, this allowed us to make best use of their version control tool. Also the platform enabled us to branch and consider several alternatives to the original game provided. Each branch was carefully considered and utilised to ensure we met all our initial objectives. Inevitably, we ran into several bugs to mitigate the impact, we would compile a list of actively known bugs and have a dedicated member of the team resolving them, this also ensured there was a clear sense of responsibility and accountability. Before each branch was merged into the main, it would be peer reviewed and assessed to ensure it aligned with our improvement specification.

Another vital aspect of the project was the use of online communication platforms, as a lot of these objectives were complete outside of timetabled sessions it was crucial we stayed connected and effective with our communication. This strengthened our group and aided us to ensure we met all our deliverables, without a strong communication skillset it would've been difficult to maintain the culture we laid out.

Requirements Change Log

The initial requirement elicitation performed by both our team and the chosen group were very similar, consisting of the same core reference point of the product brief and a constant back-and-forth between the team and the client. As such, no updates were required to the group's introduction.

Whilst the chosen group had detailed requirements and information about how they were devised, it was not intuitive to identify individual requirements, nor quick or easy to reference. As such, we decided to reformat the requirements using a tabular system. Each of their sections (User Requirements, Functional System Requirements, Non-Functional System Requirements) were changed into a table, with each requirement within having a unique ID and a description. User requirements also have a priority rating, and system requirements all reference which user requirement they contribute towards. The IDs allow for easier referencing from both within and outside of the document.

The reformatting covered the majority of changes, but some additional requirements were added to reflect the client's updated brief. These include UR_ACHIEVEMENTS (and all subsections thereof) and FR_LEADERBOARD.

Link to original: [Link](https://eng1-group9.github.io/website/Req1.pdf) (<https://eng1-group9.github.io/website/Req1.pdf>)

Link to updated: [Link](https://docs.google.com/document/d/1WCuQX2MODL0ZjRuTVhxSjDRZbJ4MMZXvePErTOG34Pc/edit?usp=sharing) (

<https://docs.google.com/document/d/1WCuQX2MODL0ZjRuTVhxSjDRZbJ4MMZXvePErTOG34Pc/edit?usp=sharing>)

Architecture Change Log

Our chosen group's architecture documentation varied significantly from ours due to a difference in design choice and system structure. However, we felt the documentation they provided sufficiently described the system they wanted to construct, and as such not many changes were required.

There were sufficiently detailed diagrams that covered all aspects of the system, both structural and behavioural. They went in depth on the tools and processes used to create each diagram, and provided links to previous iterations of the representations.

The document contained references to a formal requirements list which justified the existence of each component.

Edits to some diagrams were made (e.g. Figure 1) to reflect the client's updated product brief. Figure 9 was also added for the same purpose. Changes were made using the same tools as originally described by the chosen group (namely plantUML).

Link to original: [Link](https://eng1-group9.github.io/website/Arch1.pdf) (<https://eng1-group9.github.io/website/Arch1.pdf>)

Link to updated: [Link](https://docs.google.com/document/d/1KpV89Jej7IJLgl6ToGPJs69YzxTOW-IhDa2vIB_s08/edit?usp=sharing) (

https://docs.google.com/document/d/1KpV89Jej7IJLgl6ToGPJs69YzxTOW-IhDa2vIB_s08/edit?usp=sharing)

Method Selection and Planning Change Log

Fundamentally, our original framework very closely mimicked our chosen group's method selection embedding those core scrum principles, employing an importance of a client focused design approach. However, there was a lack of signposting alternative development methods to overcome this; we've explicitly discussed alternative methods (waterfalling) so there is a clear demonstration of consideration when applying the best methodology. Throughout the documentation there is a clear understanding of specific engineering philosophies that is primarily displayed with the explanation of Extreme Programming (XP). While our initial documentation implicitly suggested such an approach, the current version clearly demonstrates such principles and therefore requires no further change. The discussion about tools had depth and included appropriate and relevant alternatives, feeling sufficient enough to not warrant any changes.

Organisationally, the initial approach was extremely similar, working out individual qualities and assigning the most relevant roles. The current approach has lack of consideration for disagreements within the group, within our previous plan we incorporated "De Bono's Model", which we feel would be beneficial to include as it allows for designated aspects of discussion to always be present and considered. In addition to this, the current plan didn't justify why their approach was most appropriate; this has been explicitly added to ensure all criteria for assessment 1 has been achieved.

Finally, the current plan contained a decomposed, top down illustration of the project which is a core component our original lacked. Within both project planning aspects, we both illustrated our plan by the use of a Gantt chart; these effectively displayed all relevant information such as: key tasks, dates, priorities and dependencies and therefore required no changes. However, a crucial aspect which was missing from our previous documentation and has been embedded within the current - the allocation of key personnel to each task. Additionally, we lacked depth and analysis on a weekly basis, this has been carefully considered and baked into the current project plan.

Link to original: [Link](https://eng1-group9.github.io/website/Plan1.pdf) (<https://eng1-group9.github.io/website/Plan1.pdf>)

Link to updated: [Link](https://docs.google.com/document/d/1gf29F0fVtb9UxZNUDwwys8_toz4thGgRfniw2yF2VYw/edit?usp=sharing) (https://docs.google.com/document/d/1gf29F0fVtb9UxZNUDwwys8_toz4thGgRfniw2yF2VYw/edit?usp=sharing)

Risk Assessment and Mitigation Change Log

Initially, comparing our original risk assessment to our chosen groups we felt the same principles were demonstrated within the "Risk Management Process". The process in which risks were identified, deemed, and mitigated were similar enough to not require any specific changes. In some aspects the chosen groups risk analysis approach felt more polished and well defined; this is highlighted by their explicit definition of each grading terminology, giving exact percentages allows a client to precisely measure a risk compared to our arbitrary gradings given within our original risk assessment.

Notably, this current risk assessment didn't embody a clear illustration of hierarchy when determining severity; to overcome this we have implemented a colour scheme to capture different definitions and present them in a more explicit manner. Additionally, the protocol used to calculate IDs felt overwhelming, and unintuitive to streamline the process we've altered the identifications - simply labelling the risks in a sequential manner. We still felt the categorisation was important so have adopted a method in which each risk is additionally identifiable by a label: Project (Pro), Technical (Tec). These labels can be combined where necessary and enable a more precise catalogue of risks. The majority of documented risks felt extremely clear and thorough however it felt the entire scope of the project had not been captured, there was a clear prioritisation of technical risks, which is understandable, however felt there was insufficient representation of planning and management risks. To ensure all aspects of the project were correctly mitigated we have added the following risks to the risk matrix: "Insufficient planning (14)", "Poor Communication (15)". To ensure all risks were decomposed and thoroughly documented we systematically interrogated each risk. This is explicitly clear as we decided to split up the risk "TP1", the documentation of this risk felt extremely broad. Additionally, the mitigation method didn't appropriately reflect all three risks mentioned and has been updated to reflect these changes. Some of the "owners" of each risk were too broad and needed to be specified appropriately - we were unable to achieve this as we weren't familiar with the allocation of tasks within the previous group.

Finally, modifications to the presentation of the matrix felt necessary, no structural changes are to be noted however simple design choices have been altered to allow for easier comprehension and a more appealing illustration.

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